

TUFT REGISTRY STUDIES

Structural Analysis for the ZFC / IRRPROOF Engine

*Möbius Transformations as a Minimal Verified Model of
Invariant Extraction, Normal-Form Reduction, and Proof Transport*

Companion deliverables

This standalone paper is paired with a browser-based minimum-case demonstrator. The paper ingests the prior TUFT registry, Residue Thesis, framing, and IRRPROOF architecture; the demonstrator resolves those abstractions into one executable structural proof path: raw linear-fractional expression -> projective object -> invariants -> classification -> normal form -> transported conclusion.

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EmergenceByDesign

Continuing TUFT Registry Studies - Laboratory Note - July 2026

Reason for recording. This manuscript timestamps and organizes independent progress. Standard set theory, complex analysis, projective geometry, group actions, and the cited work remain attributed to their established sources. No claim of discovery, ownership, or priority is made for established results. The contribution claimed here is the declared structural-analysis architecture, its integration with the existing TUFT/Residue/IRRPROOF canon, and the companion executable demonstration.

Abstract

A proof engine that operates only on submitted syntax can verify correct deductions while repeatedly failing to recognize that differently presented problems instantiate one structural object. The present TUFT Registry Study specifies a structural-analysis layer for the ZFC-grounded IRRPROOF engine. The layer does not replace the proof kernel. It proposes typed object extraction, quotient-aware identity, invariant calculation, classification, normal-form reduction, and proof transport; each proposed structural move is compiled into a finite certificate checkable by the existing kernel.

Möbius transformations provide the minimum complete case. A raw expression $T(z)=(az+b)/(cz+d)$, $ad-bc \neq 0$, may be read as a rational function, a projective matrix class in $PGL(2,C)$, an automorphism of the Riemann sphere, a generalized-circle-preserving conformal map, a cross-ratio-preserving action, and a conjugacy class determined by fixed-point and multiplier data. These views are not competing descriptions. They form a registry of mutually translatable witnesses of one object. The paper formalizes that registry, distinguishes coordinate artifact from conjugacy-invariant structure, and defines a certificate pipeline from raw syntax to a canonical proof route.

The architecture ingests prior TUFT claims that the registry precedes the operator, the Residue Thesis distinction $O=A+I_R$, the Geometry of Framing's requirement that every theorem remains indexed to its framing discipline, and IRRPROOF's full-object-chain/certified-truncation model. The companion HTML demonstrator implements exact formula generation, numerical complex arithmetic for real-coefficient examples, projective normalization, fixed-point extraction, scale-invariant trace classification, conjugacy reduction summaries, cross-ratio verification, matrix composition, and a human-readable certificate ledger. It is intentionally a minimum case rather than a universal prover.

1. Position within the existing canon

The present work should be read as a continuation of TUFT Registry Studies rather than an independent replacement framework. The earlier registry work fixes the carrier before selecting an operator and treats Beltrami, Dirac, and Hamiltonian spectra as different readings of a common relational substrate. The Residue Thesis identifies structure with the obstruction that survives admissible re-description and requires every irreducibility claim to remain indexed to its representation class. The Geometry of Framing makes explicit that objects, witnesses, transformations, and stopping conditions acquire meaning only inside a declared framing discipline. IRRPROOF then supplies the engineering consequence: preserve the full certified object chain while compiling smaller domain engines as dependency-closed, kernel-verifiable truncations.

New step

Structural analysis turns those principles into a proof-search mechanism. It asks: What object does the submitted expression present? Which changes are admissible re-descriptions? Which quantities survive them? Which normal form represents the resulting structural class? Which theorem already proved for that normal form can be transported back?

1.1 Tier discipline

Tier	Status	Content	Claim boundary
I	Established host mathematics	ZFC encoding; complex field; Riemann sphere; projective matrices; Möbius maps; cross	Cited and re-proved only where needed by the engine.

Tier	Status	Content	Claim boundary
		ratios; generalized circles; conjugacy normal forms.	
II	Canonical TUFT/Residue reading	Carrier-before-operator; structural invariant as re-description residue; explicit framing; registry of equivalent witnesses.	Established within the declared corpus frame.
III	Present frontier construction	Automated object extraction, strategy selection, structural proof certificates, residue telemetry, registry learning.	Architecture and minimum implementation, not a claim of universal structural recognition.

2. Why syntax-only proof search remains incomplete as an engineering method

A sound ZFC kernel can verify a derivation without knowing that the derivation was unnecessarily long. It can also fail to locate a proof because the target theorem was stored under a different presentation. This is not logical incompleteness in the Gödelian sense. It is representational incompleteness of the search interface.

submitted syntax -> local rewrites -> candidate proof

Structural analysis inserts an intermediate object layer:

submitted syntax -> typed object -> admissible quotients -> invariants -> normal form -> proof transport

The kernel remains small and authoritative. The structural analyzer remains replaceable and untrusted. It earns every optimization by emitting a certificate whose primitive expansion the kernel can check.

2.1 Structural compression

Suppose a registry contains separate proofs P_i for objects x_i that later become recognized as instances of one normal-form theorem P^* . The registry can retain:

- **one general theorem**, proved for the canonical representative;
- **one classifier**, showing that each x_i belongs to the represented structural class;
- **one transport witness per instance**, showing how P^* returns to x_i ;
- **the original proofs**, until migration and equivalence certificates permit archival compression.

No theorem disappears. Independent descriptions become conditional on a deeper shared substrate.

3. The minimum structural object: Möbius transformation

Let $\hat{C} = C \cup \{\infty\}$. A Möbius transformation is represented by

$$T(z) = (az + b)/(cz + d), \quad ad - bc \neq 0.$$

When $c \neq 0$, $T(-d/c) = \infty$ and $T(\infty) = a/c$. When $c = 0$, $T(\infty) = \infty$. The derivative on the finite domain is

$$T'(z) = (ad - bc)/(cz + d)^2,$$

so the transformation is conformal wherever finite. Every such transformation is a bijection of $\hat{C}$, and composition corresponds to matrix multiplication.

3.1 Projective representation

$$M_T = \begin{bmatrix} a & b \\ c & d \end{bmatrix}, \quad \det(M_T) \neq 0.$$

For every nonzero scalar s , M_T and $s M_T$ define the same transformation. Therefore the structural identity lives in $\text{PGL}(2, \mathbb{C})$, not in the raw coefficient tuple.

$$[M] = \{s M : s \in \mathbb{C}^*, s \neq 0\}.$$

This quotient is the first exact example of the Residue Thesis inside the engine: coefficient scale belongs to presentation artifact; projective action survives.

3.2 Matrix composition and inverse

$$T_M \circ T_N = T_{(MN)}, \quad T_M^{-1} = T_{(M^{-1})}.$$

A long nested rational expression can therefore be compiled into matrix multiplication, checked, and converted back only when a final formula is required. The structural route reduces expression swell and exposes group relations.

3.3 Extended-plane and generalized-circle action

The Riemann sphere supplies the natural total domain because poles become ordinary images of finite points and infinity becomes an admissible point. Möbius transformations send generalized circles - Euclidean circles or Euclidean lines completed by infinity - to generalized circles. The engine should store this as a type-preservation theorem rather than repeatedly eliminate coordinates.

4. Cross ratio as a complete local relational invariant

$$[z, z_1; z_2, z_3] = ((z - z_1)(z_2 - z_3)) / ((z - z_3)(z_2 - z_1)).$$

For distinct marked points, Möbius transformations preserve the cross ratio. Conversely, three distinct source points and three distinct target points determine a unique Möbius transformation. This is a near-ideal proof-engine macro because it replaces coefficient solving with one relational equation.

Theorem 2 (Three-point determination).

Given pairwise distinct z_1, z_2, z_3 in $\hat{C}$ and pairwise distinct w_1, w_2, w_3 in $\hat{C}$, there exists a unique T in $\text{PGL}(2, \mathbb{C})$ such that $T(z_i) = w_i$ for $i=1, 2, 3$.

Proof sketch. Choose projective maps C_z and C_w sending the two triples to a common canonical triple, for example $(0, 1, \infty)$. Then $T = C_w^{-1} \circ C_z$. Existence follows by construction. If S has the same three values, $C_w \circ S \circ C_z^{-1}$ fixes $0, 1, \infty$; the corresponding matrix is scalar, hence the identity in $\text{PGL}(2, \mathbb{C})$. QED.

4.1 Certificate form

The structural analyzer need not be trusted to solve the mapping correctly. It emits:

- **distinctness witnesses**, for both triples;
- **a candidate matrix M** , with nonzero determinant;
- **three evaluation identities**, $T(z_i) = w_i$;
- **a uniqueness citation**, to the three-point theorem.

5. Fixed points, multipliers, and conjugacy

Finite fixed points solve

$$c z^2 + (d-a) z - b = 0.$$

Counting infinity, every nonidentity Möbius transformation has either two distinct fixed points or one double fixed point.

5.1 Distinct fixed points

If $p \neq q$ are fixed, the conjugator

$$C(z) = (z-p)/(z-q)$$

sends p to 0 and q to infinity. Therefore

$$C \circ T \circ C^{-1}(z) = k z$$

for some $k \neq 0, 1$. The multiplier k determines the qualitative conjugacy type, up to inversion when the fixed-point ordering is swapped.

5.2 Double fixed point

If T has one double fixed point p and is not the identity, a conjugator sending p to infinity reduces T to

$$z \rightarrow z + \tau, \quad \tau \neq 0.$$

A further scaling conjugacy reduces every nontrivial translation to $z \rightarrow z+1$. Thus the parabolic class has a single canonical normal form over $\mathbb{C}$.

5.3 Scale-invariant trace classifier

Raw trace is not projectively invariant. The engine should first compute

$$\chi(M) = \text{tr}(M)^2 / \det(M).$$

This remains unchanged under $M \rightarrow sM$. For determinant-one representatives, $\chi = \text{tr}(M)^2$. Classification over $\text{PSL}(2, \mathbb{C})$ can then be expressed without selecting a square-root branch:

Condition	Structural class	Canonical form	Fixed-point structure
$\chi = 4$, nonidentity	Parabolic	$z+1$	One double fixed point
χ real and $0 < \chi < 4$	Elliptic	$\exp(i \theta) z$	Two distinct fixed points
χ real and $\chi > 4$	Hyperbolic	$k z, k > 0, k \neq 1$	Two distinct fixed points
χ not real, or complex convention outside prior cases	Loxodromic	$k z$ with dilation and rotation	Two distinct fixed points

Classification conventions vary, especially concerning negative real multipliers and whether hyperbolic is treated as a loxodromic subtype. The engine must store the convention as part of the framing discipline instead of silently mixing sources.

6. Structural residue for Möbius objects

Let R_{proj} permit nonzero scalar changes of matrix representative. Let R_{conj} additionally permit conjugation by $\text{PGL}(2, \mathbb{C})$. For an obstruction or description-cost functional O , define

$$I_R(T; \text{target}) = \inf_{r \in R(T)} O(r, T; \text{target}), \quad O = A + I_R.$$

Under R_{proj} , the exact coefficient scale is removable artifact. Under R_{conj} , fixed-point locations are removable, but fixed-point multiplicity, multiplier class, and conjugacy type persist. The index matters: a quantity may be structural under projective rescaling yet removable under conjugation.

Feature	Invariant under projective rescaling?	Invariant under conjugation?	Engine role
Coefficient tuple (a,b,c,d)	No	No	Raw presentation
Projective matrix class	Yes	No	Object identity
determinant zero/nonzero	Yes as a predicate	Yes	Validity
Fixed-point locations	Yes	No	Construct conjugator
Fixed-point multiplicity	Yes	Yes	Class invariant
$\chi = \text{tr}^2 / \det$	Yes	Yes	Classifier
Cross ratio of marked points	Yes	Yes under simultaneous action	Relational invariant
Multiplier up to inversion	Yes	Yes	Normal-form parameter
Generalized-circle incidence	Yes	Yes	Geometric transport

7. ZFC-grounded object model

The structural module can be conservative over ZFC. It requires no new foundational primitive. Each high-level object expands into ordinary sets, functions, relations, and finite sequences. A practical implementation may use typed terms while retaining a verified erasure into the kernel's set-theoretic language.

7.1 Foundational dependencies

1. Construct natural numbers, integers, rationals, reals, and complex numbers in the selected ZFC library.
2. Define $\hat{C}$ as a tagged disjoint union C plus one infinity element.
3. Define 2×2 matrices as finite functions on $\{0,1\} \times \{0,1\}$.
4. Define $GL(2,C)$, scalar equivalence, $PGL(2,C)$, and the induced action on projective one-space.
5. Prove equivalence between projective-line action and the extended fractional formula.
6. Define generalized circles, cross ratios with infinity-aware clauses, fixed points, composition, inverse, and conjugation.
7. Register theorem objects with explicit dependencies and framing conventions.

7.2 Trusted and untrusted components

Component	Trusted?	Function	Failure consequence
ZFC proof kernel	Yes	Checks primitive derivations	Unsoundness if defective
Parser/elaborator	No	Turns submitted syntax into typed terms	Rejected or wrong candidate, caught by checking

Component	Trusted?	Function	Failure consequence
Structural extractor	No	Proposes matrix, invariants, fixed points	Certificate fails
Strategy selector	No	Chooses cross-ratio, conjugacy, or matrix route	Search inefficiency only
Computer algebra helper	No	Factors polynomials and simplifies identities	Certificate fails
Registry/index	No for truth, yes for availability	Finds reusable theorems	Missed optimization, not false proof

8. Structural certificate calculus

A structural proof certificate is a finite object that justifies the transport from submitted presentation to a theorem-bearing normal form.

Definition 3 (Möbius structural certificate).

A certificate is a tuple $(E, M, V, I, K, C, N, P, \Pi)$, where E is the parsed expression; M a projective matrix witness; V a nondegeneracy proof; I a vector of invariants; K a classification; C a conjugator or three-point map; N a normal form; P a theorem proved for N ; and Π a transport derivation.

8.1 Verification obligations

8. Expression equivalence: prove $E(z) = T_M(z)$ on $C\text{-hat}$.
9. Validity: prove $\det(M) \neq 0$.
10. Quotient discipline: prove every normalization preserves $[M]$.
11. Invariant correctness: prove the fixed-point polynomial, chi value, and cross-ratio claims.
12. Class membership: derive the stated class from the registered convention.
13. Normal-form identity: prove $C \circ T_M \circ C^{-1} = N$.
14. Normal-form theorem: reference or check $P(N)$.
15. Transport: prove conjugacy or naturality carries $P(N)$ back to $P(T_M)$.
16. Scope disclosure: record all side conditions and the representation class under which the reduction holds.

8.2 Generic transport theorem

Theorem 4 (Conjugacy transport).

Let G act on a class X , let P be a predicate invariant under the action, and let $y = g.x$. If $P(x)$, then $P(y)$.

Proof. Invariance gives $P(x)$ iff $P(g.x)$. Substitute $y = g.x$. QED.

The useful work lies in registering exactly which predicates are conjugacy invariant. Orbit convergence to a named coordinate point is not invariant without transporting the point; number of fixed points, periodicity, and generalized-circle incidence are invariant in the appropriate relational form.

9. Proof-search strategies compiled from structural signatures

Detected signature	Preferred structural route	Canonical object	Typical theorem family
Three marked	Cross-ratio / canonical	Unique three-point map	Conformal domain

Detected signature	Preferred structural route	Canonical object	Typical theorem family
source/target points	triple		normalization
Long composition	Projective matrix product	Single PGL element	Identity, inverse, word reduction
Iteration T^n	Fixed points + conjugacy	$k^n z$ or $z+n$	Orbit and stability analysis
Circle/line boundary	Generalized-circle transport	Real line or unit circle	Domain mapping
Equality of transformations	Projective matrix proportionality or three-point agreement	PGL identity	Extensional equality
Classification request	chi + fixed-point discriminant	elliptic/parabolic/hyperbolic/loxodromic	Qualitative dynamics
Large rational expression	Matrix extraction and canonicalization	Normalized projective representative	Simplification and reuse

10. IRRPROOF integration

The full IRRPROOF object chain should store more than the final theorem. For each successful reduction it should preserve the submitted expression, typed object, projective representative, invariant signature, chosen normal form, conjugator, proof transport, primitive expansion, and measured proof-cost change.

10.1 Structural signature index

$\text{sig}(T) = (\text{object_type}, \text{coefficient_field}, \text{quotient_class}, \text{fixed_point_type}, \text{chi}, \text{multiplier_class}, \text{preserved_relations})$.

The index lets the engine search by structure rather than vocabulary. A theorem proved for one ugly coefficient tuple becomes discoverable for every conjugate instance.

10.2 Failed routes as residue telemetry

A failed cross-ratio construction, unresolved factorization, singular normalization, or unproved side condition should remain in the registry with its declared resource and representation bounds. It may later become solvable after a stronger algebraic-number package or a larger theorem library is installed.

10.3 Domain truncation

A Möbius domain engine can be compiled as a certified truncation containing only the complex field, projective 2×2 matrices, $\hat{C}$, cross-ratio, generalized-circle, fixed-point, conjugacy, and certificate theorems. The complete federation retains the broader ZFC and TUFT registry so that future bridges remain available.

11. Minimum-case companion demonstrator

The companion HTML is not presented as a proof assistant or as a substitute for the ZFC kernel. It is an executable specification of the structural pipeline. It accepts real coefficients a, b, c, d ; rejects singular matrices; extracts the projective object; computes the pole, infinity image, fixed points, discriminant, chi, classification,

and a normal-form summary; compares scalar-equivalent representatives; composes two transformations by matrix multiplication; verifies cross-ratio preservation numerically on a user-selected point quadruple; and emits a certificate ledger.

11.1 Demonstrator resolution path

raw coefficients -> PGL class -> invariant signature -> class -> normal form -> certificate ledger

The interface intentionally shows both structure and scope. Numerical cross-ratio checks are labeled demonstrations, not kernel proofs. Exact identities used for classification are shown symbolically where the real-coefficient minimum case permits.

11.2 Included canonical examples

- **Hyperbolic:** $T(z)=(2z+1)/(z+1)$, with $\chi=9$ and fixed points $(1 \pm \sqrt{5})/2$.
- **Parabolic:** $T(z)=z+1$, with $\chi=4$ and one double fixed point at infinity.
- **Elliptic:** $T(z)=-1/z$, with $\chi=0$ and fixed points $\pm i$.
- **Projective equality:** M and $2M$ produce identical evaluations.
- **Composition:** matrix product is compared against nested evaluation.

12. Correctness cautions inherited from the source material

The linked complex-analysis page is useful as an instructional source, but the formal engine should repair several points before ingestion.

- **Pole sign:** the finite pole is $-d/c$, not d/c . The source's extended-plane formula is correct, while one prose domain sentence drops the minus sign.
- **Dynamics language:** a single hyperbolic or loxodromic Möbius transformation has highly organized fixed-point dynamics. Calling it generically chaotic on the entire sphere is too broad. Chaos becomes natural for non-elementary groups, semigroups, or induced actions on limit sets under precise definitions.
- **Trace convention:** trace classification requires determinant normalization or the scale-invariant $\chi=\text{tr}^2/\det$. Raw trace is not a PGL invariant.
- **Elliptic periodicity:** elliptic does not imply finite periodic order unless the rotation angle is a rational multiple of 2π .

13. Worked structural traversal

Consider

$$T(z) = (2z+1)/(z+1), \quad M = \begin{bmatrix} 2 & 1 \\ 1 & 1 \end{bmatrix}.$$

- Validity: $\det(M)=1$, so T is a Möbius transformation.
- Projective identity: any nonzero scalar multiple represents the same T .
- Fixed-point equation: $z^2-z-1=0$, giving $p=(1+\sqrt{5})/2$ and $q=(1-\sqrt{5})/2$.
- Scale-invariant classifier: $\chi=\text{tr}(M)^2/\det(M)=9$.
- Under the registered convention, $\chi>4$ real, so T is hyperbolic.
- Choose $C(z)=(z-p)/(z-q)$. Then C sends p to 0 and q to infinity.
- The conjugated map fixes 0 and infinity, hence has form kz .
- The multiplier can be obtained from $T'(p)=1/(p+1)^2$; exchanging p and q replaces k by $1/k$.
- Iteration reduces to $C(T^n(z))=k^n C(z)$.
- Any conjugacy-invariant theorem proved for kz is transported back to T .

This traversal shows why the structural layer matters. Direct expansion of T^n creates rapidly growing rational expressions. Conjugacy turns the same family into one exponentiation.

14. Generalization beyond Möbius transformations

The minimum case is selected because every architectural layer is visible and finitely checkable. The same pattern extends:

Domain	Quotient / equivalence	Invariant signature	Normal form
Matrices	Similarity	Characteristic/minimal polynomial, Jordan data	Jordan or rational canonical form
Quadratic forms	Congruence	Rank, signature, discriminant	Diagonal form
Finite groups	Isomorphism	Order spectrum, composition structure	Canonical labeled presentation
Graphs	Graph isomorphism	Canonical labeling, spectra, motifs	Canonical graph
Knots	Ambient isotopy	Polynomial and homological invariants	Certified diagram class
Differential operators	Unitary equivalence / conjugacy	Spectrum, spectral measure, index	Model operator
Formal theories	Interpretation / definitional equivalence	Consequence and translation invariants	Canonical interface theory

No single invariant set is complete in every domain. The engine must distinguish a classifier useful for search from a complete invariant sufficient for identity.

15. Frontier research program

- **Certified canonicalization over algebraic numbers:** replace floating root selection with exact algebraic-number witnesses.
- **Infinity-aware cross-ratio library:** formalize all marked-point cases without ad hoc limit prose.
- **Predicate transport registry:** tag theorems by invariance under projective rescaling, conjugacy, anti-conformal maps, and field embeddings.
- **Proof-cost telemetry:** measure primitive proof length, search time, dependency fan-out, and reuse before and after structural reduction.
- **Counterexample generation:** use normal forms to identify when an apparently invariant statement actually depends on coordinates.
- **Automated frame disclosure:** make the engine print the selected equivalence relation and classification convention with every result.
- **TUFT bridge:** use the same carrier/operator distinction for projective actions, spectral operators, and calibrated physical registries.
- **IRRPROOF federation:** share structural certificates and canonical theorem objects across replicas while compiling minimal domain analyzers.

16. Claims, non-claims, and stopping state

16.1 Claims

- **C1.** Möbius transformations furnish a complete minimum example of quotient-aware structural proof analysis.

- **C2.** Projective identity, cross-ratio invariance, fixed-point classification, and conjugacy normal forms can be compiled into finite ZFC-checkable certificates.
- **C3.** The architecture implements the prior TUFT/Residue/Framing distinction between presentation artifact and persistent structure.
- **C4.** IRRPROOF can store and reuse normal-form proofs without enlarging the trusted kernel.
- **C5.** The companion demonstrator resolves the architecture into an inspectable executable traversal.

16.2 Non-claims

- **N1.** The demonstrator is not a complete complex arithmetic system or formal ZFC verifier.
- **N2.** Numerical cross-ratio agreement is not a proof certificate.
- **N3.** The chosen invariant signature is not a complete invariant for arbitrary mathematical objects.
- **N4.** Structural recognition does not evade undecidability or incompleteness.
- **N5.** No ownership claim is made over Möbius theory, projective geometry, or established proof-assistant methods.

16.3 Stopping state

This paper reaches its stated stopping state when the architecture, kernel boundary, certificate obligations, minimum mathematical domain, demonstrator behavior, and generalization path have all been specified. The next legitimate step is implementation against the actual ZFC proof-object format and empirical measurement of proof compression.

17. Conclusion

A ZFC proof engine becomes structurally stronger not by weakening formal verification but by inserting a richer, certificate-producing search layer above it. Möbius transformations expose the full pattern in its smallest coherent form: one expression becomes a projective object; projective rescaling removes coefficient artifact; cross ratios encode relational action; fixed points and chi classify the object; conjugacy supplies a normal form; and theorem transport returns the result to the submitted coordinates.

This is the operational resolution of the prior TUFT registry work. The registry is not merely a collection of formulas. It is a network of admissible witnesses and transports. The Residue Thesis tells the engine what to preserve; framing discipline tells it what equivalence relation has been declared; IRRPROOF tells it how to retain certificates, failures, and reusable compressed proofs. The companion demonstrator makes that architecture visible in one finite traversal.

Final formulation

Structural analysis for IRRPROOF is an untrusted search-and-canonicalization layer that converts raw ZFC expressions into typed quotient objects, computes declared invariants, reduces them to theorem-bearing normal forms, and emits primitive certificates for kernel verification. Möbius transformations provide the first complete executable model.

Appendix A. Certificate schema

```
{
  "object_type": "MobiusTransformation",
  "expression": "(a*z+b)/(c*z+d)",
  "matrix": [[a,b],[c,d]],
  "validity": "a*d-b*c != 0",
  "quotient": "M ~ scalar*M",
  "invariants": {
```

```

"fixed_point_polynomial": "c*z^2+(d-a)*z-b",
"chi": "(a+d)^2/(a*d-b*c)",
"fixed_point_multiplicity": "...",
"cross_ratio_preserved": true
},
"classification_convention": "...",
"class": "...",
"conjugator": "...",
"normal_form": "...",
"transported_theorem": "...",
"primitive_proof_ref": "..."
}

```

Appendix B. Demonstrator test suite

Test	Input	Expected structural result	Purpose
T1	2,1,1,1	det=1; chi=9; hyperbolic	Classification
T2	1,1,0,1	chi=4; parabolic; fixed infinity	Infinity handling
T3	0,-1,1,0	chi=0; elliptic; fixed +/-i	Complex fixed points
T4	M and 2M	same evaluations	Projective quotient
T5	A o B	matrix product equals nested action	Composition
T6	four finite points	cross ratios agree within tolerance	Relational invariant
T7	ad-bc=0	reject object	Validity boundary

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